

Wireless Sensors Network

Sensor Networking -- Vision

WSN Application Examples

- **Emergency operations**
 - Drop sensor nodes from an aircraft over a wildfire
 - Each node measures temperature
 - Derive a "temperature map"
- **Habitat monitoring**
 - Use sensor nodes to observe wildlife
 - E.g., Great Duck Island, ZebraNet
- **Precision agriculture**
 - Bring out fertilizer/pesticides/irrigation only where needed
- **Logistics**
 - Equip goods (parcels, containers) with a sensor node
 - Track their whereabouts – total asset management.



WSN Application Scenarios

■ Home automation and health care

- Smart environment (smart sensor nodes and actuators in appliances learn to provide needed service)
- Post-operative or intensive care (telemonitoring of physiologic data)
- Long-term surveillance of chronically ill patients or the elderly (tracking and monitoring)

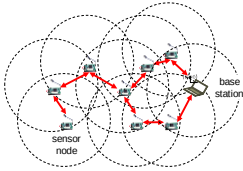
The diagram illustrates a smart home environment with various sensors and actuators. A central dashed circle labeled 'High Resolving Application' contains a floor plan of a house. Sensors and actuators are placed throughout the house: Temperature Sensor (top left), Light Sensor (top right), Motion Sensor (bottom left), Humidity Sensor (bottom right), and a Health Care Monitor (center). A central 'Internet / Telephony Network' is connected to the house. Outside the house, there are icons for 'Family', 'Pet', 'Provision' (a plus sign), and 'Hospital' (a cross).

A photograph of a physical model of a smart home environment. It shows a small, rectangular, light-colored structure with a flat top surface. The structure is divided into several sections, some of which are covered with blue material. The model is placed on a dark surface, and a person's hand is visible in the bottom right corner, indicating its scale.

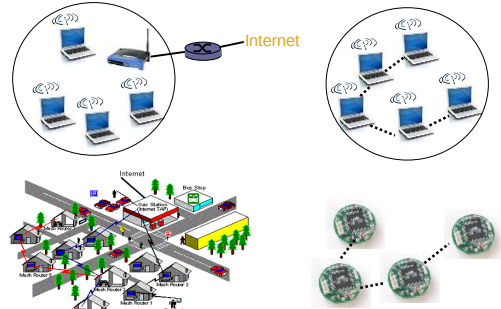
Wireless Sensor Networks

■ Multiple roles can be distinguished

- *Sensors* – measure physical phenomena, sources of measurement data
- *Base stations* – analyze and post-process data, sinks for measurement data
- *Actuators* – perform actuation in response to received data
- *Processing elements* – pre-processing of transmitted data



Wireless networks



- Wireless LAN, ad-hoc network, wireless mesh, sensor network

Sensor Hardware

■ *Sensor Node* contains

1. Sensor(s) to convert a energy form to an electrical impulse
 - e.g., to measure temperature
2. Microprocessor
3. Communications link e.g., wireless
4. Power source e.g., battery

AdHoc and Sensors ...

■ Ad Hoc and Sensor Networks are both multi-hop wireless architectures

- Thereby shares several technical issues and challenges

However, key differences exist

- Energy constraint in sensor networks
- Traffic models and characteristics
- Other issues like coverage, fault-tolerance, etc.

Operation and Maintenance

- Feasible and/or practical to maintain sensor nodes?
 - E.g., to replace batteries?
 - Or: unattended operation?
 - Impossible but not relevant? Mission lifetime might be very small
- Energy supply?
 - Limited from point of deployment?
 - Some form of recharging, energy source from environment?
 - E.g., solar cells